Pauli Gates

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$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \& \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

By applying X-gate:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0+0 \\ 1+0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$X|1\rangle = \begin{bmatrix} 0+1 \\ 0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X|1\rangle = |0\rangle$$

By applying Y-gate:

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Y|0\rangle = \begin{bmatrix} 0+0 \\ i+0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix}$$

$$Y|0\rangle = i\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Y|0\rangle = i|1\rangle$$

$$Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Y|1\rangle = \begin{bmatrix} 0+-i \\ 0+0 \end{bmatrix}$$

$$Y|1\rangle = \begin{bmatrix} -i \\ 0 \end{bmatrix}$$

$$Y|1\rangle = -i\begin{bmatrix} 1 \\ 0 \end{bmatrix} = -i|0\rangle$$

By applying Z-gate:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Z|0\rangle = \begin{bmatrix} 1+0 \\ 0+0 \end{bmatrix}$$

$$Z|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Z|1\rangle = \begin{bmatrix} 0+0 \\ 0+-1 \end{bmatrix}$$

$$Z|1\rangle = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$Z|1\rangle = -1\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Z|1\rangle = -1|1\rangle$$